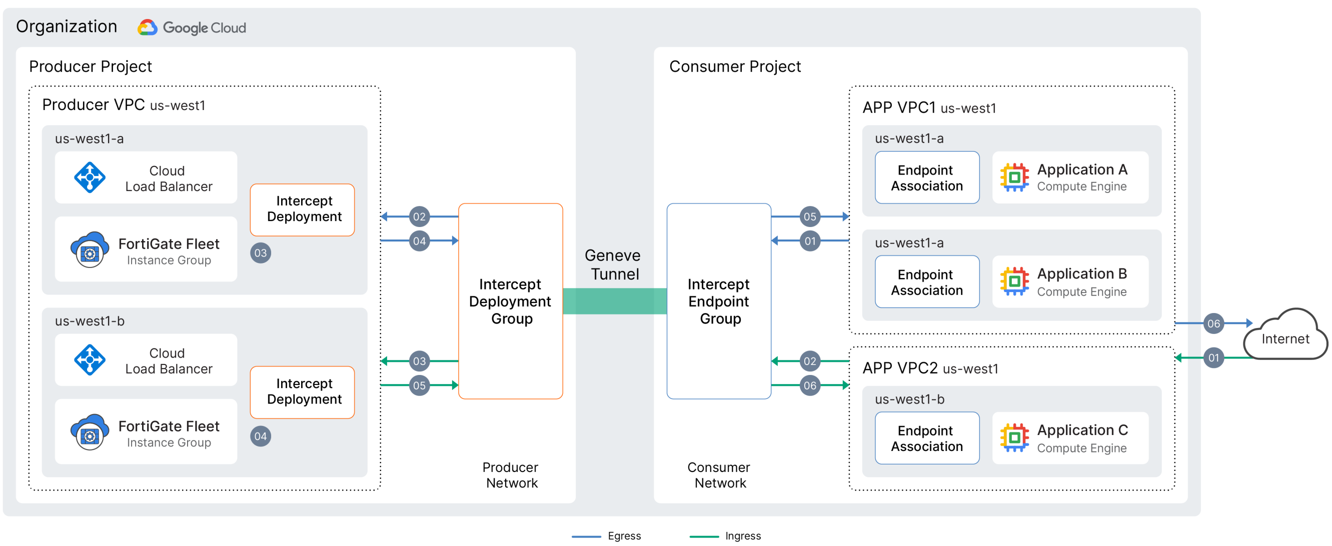
Network Security Integration inband deployment with FortiGate-VM

FortiGate-VM on GCP supports the GCP Network Security Integration inband (NSI inband) feature designed for packet inspection of workloads running within GCP. NSI inband is a technology that enables transparent interception, inspection, and reinjection of customer traffic using simple GCP firewall policies and rules as well as FortiGate's robust unified threat management features.

This guide focuses on a single region while using multiple zones within that region.

The following shows the example topology for this deployment:



The following summarizes the VPCs in this example:

VPC	Description
Producer	Comprised of a Traffic VPC and a Management VPC: • Traffic VPC or network is used to receive traffic for inspection and send traffic after inspection back into the GCP network. • Management VPC or network is used to allow connection to the FortiGate-VM for management.
Consumer	Comprised of an App VPC1 and an App VPC2: • Intercept Endpoint Group (GWLB endpoints in web VPCs) is a project level construct and can be attached to multiple VPCs in multiple regions. • Intercept Endpoint Group Association, this configuration associates the Intercept Endpoint group to the web VPC and can be attached to multiple VPCs in the region. • Security Profile Group and Security profile are also an org level construct and are associated with Intercept Endpoint Group.

VPC	Description
	• Network Firewall Policy (Project level) are associated with App VPCs in this example and introduce a new action “Proceed to L7 inspection” in GCP firewall rule configuration.

This guide reference the following VPCs, regions, and zones. These can and should be changed to fit your environment.

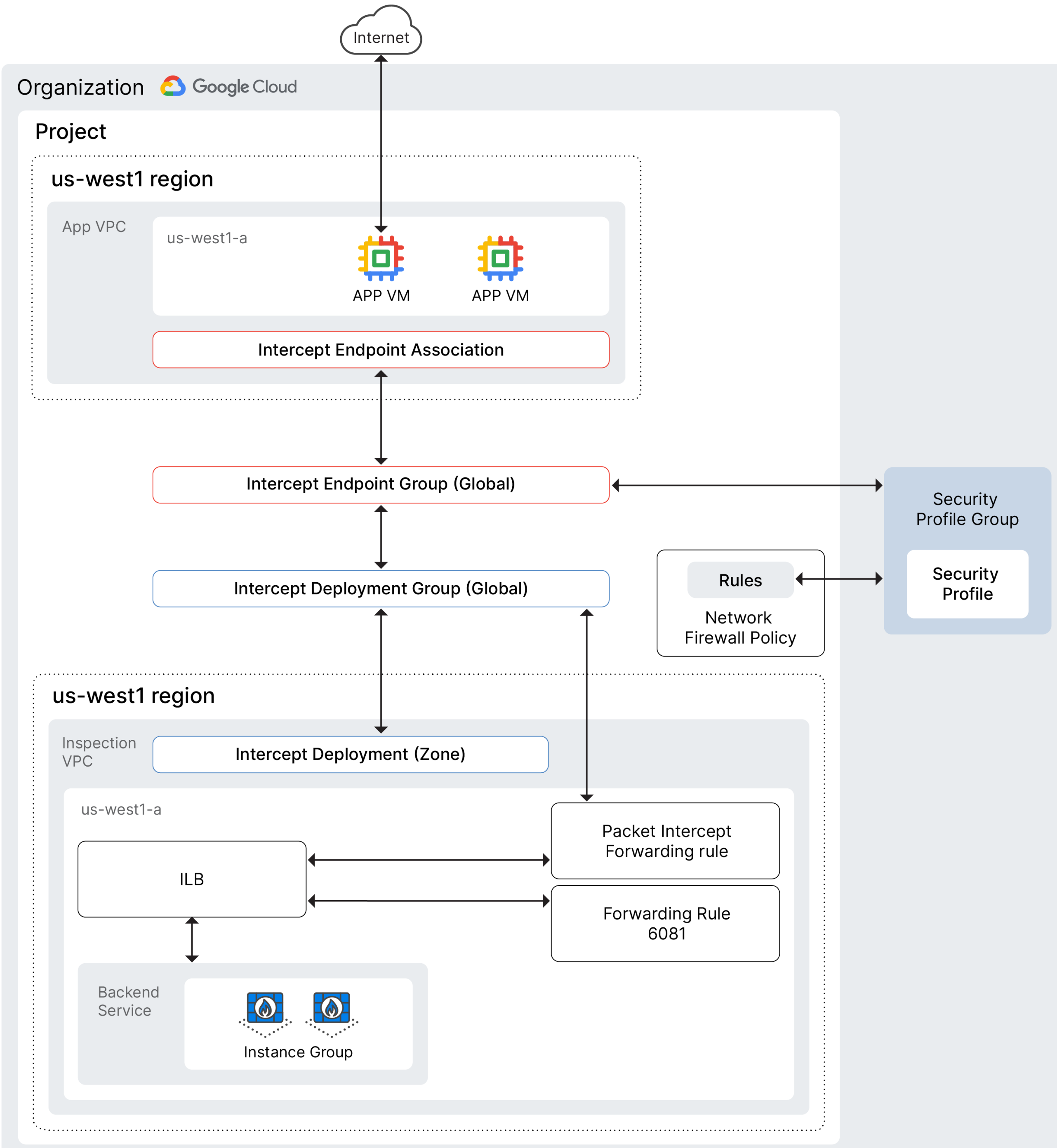
VPC type	Description
Inspection (traffic)	Where the to be inspected traffic will flow through to and from the FortiGates
Inspection (management)	Where the management console access traffic flows to manage the FortiGates.
Application	Publicly exposed workload that generates the traffic to be inspected.

The region that was used for this document was us-west1 and the zones are us-west-1a and us-west-1b.

This guide and example will assume you have gcloud installed and configured for your environment, with the appropriate permissions to create VPC networks, managed instances groups, and enabling the Network Security Integration feature.

Configuring GCP resources

The NSI inband feature will send packets to the FortiGates for inspection and then those same packets will be delivered back into the GCP networks to complete the delivery to the destination network.



To create the VPCs and subnets:

This example creates new VPCs to run the webserver that generate the traffic for inspection, the VPC that the FortiGate-VMs will use to receive the traffic for inspection, and the management VPC that will be used to manage the FortiGate-VMs.

To create new VPCs and networks, for example use the following Gcloud commands in the regions as you need to fit your environment. For this example, we chose US-WEST1. For multiregion examples see Appendix on page 145.

1. Create the data or traffic inspection VPC and subnets:

```
gcloud compute networks create fgt-nsi-ib-new \
    --subnet-mode=custom \
    --description "NIS data or traffic VPC network with regional subnets"
```

2. Create the inspection subnet in us-west1:

```
gcloud compute networks subnets create fgt-nsi-west \
    --network=fgt-nsi-ib-new \
    --region=us-west1 \
    --range=10.50.160.0/24 \
    --enable-private-ip-google-access \
    --description="Data or Traffic inspection Subnet in us-west1"
```

3. Create firewall rules for Geneve traffic:
   a. Create a firewall rule to allow all ingress Geneve traffic:

```
gcloud compute firewall-rules create fgt-nsi-allow-all-in \
  --network=fgt-nsi-ib-new \
  --allow=tcp:0-65535,udp:0-65535,icmp \
  --source-ranges=0.0.0.0/0 \
  --description="Allow all incoming data traffic for inspection"
```

   b. Create a firewall rule to allow all egress Geneve traffic:

```
gcloud compute firewall-rules create fgt-nsi-allow-all-egr \
  --network=fgt-nsi-ib-new \
  --allow=tcp:0-65535,udp:0-65535,icmp \
  --destination-ranges=0.0.0.0/0 \
  --direction EGRESS \
  --description="Allow all outgoing traffic for inspection"
```

4. Create the FortiGate management VPC with subnets:

```
gcloud compute networks create fgt-nsi-ib-new-mgmt \
  --subnet-mode=custom \
  --description="FortiGate management VPC network with regional subnets"
```

5. Create the management subnet in us-west1:

```
gcloud compute networks subnets create fgt-nsi1-mgmt-west \
    --network=fgt-nsi-ib-new-mgmt \
    --region=us-west1 \
    --range=10.50.180.0/24 \
    --enable-private-ip-google-access \
    --description="FortiGate management Subnet in us-west1"
```

6. Create firewall rules for management traffic:
   a. Create a firewall rule to allow all ingress management traffic:

```
gcloud compute firewall-rules create fgt-nsi-allow-all-ing1 \
    --network=fgt-nsi-ib-new-mgmt \
    --allow=tcp:0-65535,udp:0-65535,icmp \
    --source-ranges=0.0.0.0/0 \
    --description="FortiGate management allow all incoming traffic"
```

b. Create a firewall rule to allow all egress management traffic:

```
gcloud compute firewall-rules create fgt-nsi-allow-all-egr1 \
    --network=fgt-nsi-ib-new-mgmt \
    --allow=tcp:0-65535,udp:0-65535,icmp \
    --direction=EGRESS \
    --destination-ranges=0.0.0.0/0 \
    --description="FortiGate management allow all outgoing traffic"
```

7. Create the web VPC and subnets:

```
gcloud compute networks create fgt-nsi-ib-new-web \
    --subnet-mode=custom \
    --description="Public Web VPC network with regional subnets"
```

8. Create a subnet in us-west1:

```
gcloud compute networks subnets create fgt-nsi-web1-west \
    --network=fgt-nsi-ib-new-web \
    --region=us-west1 \
    --range=10.12.0.0/24 \
    --enable-private-ip-google-access \
    --description="Public Web Subnet in us-west1"
```

To create the FortiGate instance template:

```
gcloud compute instance-templates create fgt-nsi762-v5 \
  --project=<your project id> \
  --machine-type=c4-standard-4 \
  --network-interface=network-tier=PREMIUM,stack-type=IPV4_ONLY,subnet=fgt-nsi-west \
  --add-network-interface=network-tier=PREMIUM,stack-type=IPV4_ONLY,subnet=fgt-nsi1-mgmt-
west \
  --region=us-west1 \
  --create-disk=auto-delete=yes,boot=yes,device-name=instance-template-
2,image=projects/fortigcp-project-001/global/images/fortinet-fgtondemand-763-20250423-001-w-
license,mode=rw,size=10,type=pd-balanced
```

To create a health check:

This health check will be referenced throughout this document. Edit GCP CLI configuration examples to fit your environment.

```
gcloud compute health-checks create http doc-nsi-hc
```

To create the FortiGate-VM MIG:

This step references the previously created instance template to create a FortiGate-VM managed instance group (MIG).

```
gcloud compute instance-groups managed create fgt-nsi-us-west1-multi-ig \
  --project=<your project id> \
  --template=projects/<your project id>/global/instanceTemplates/fgt-nsi762-v5 \
  --size=3 \
  --zones=us-west1-a,us-west1-b
```

To create a backend service:

```
gcloud compute backend-services create fgt-nsi-us-west1-lb \
    --load-balancing-scheme=INTERNAL \
    --region=us-west1 \
    --health-checks= doc-nsi-hc \
    --protocol=UDP
```

To add the MIG to the backend service:

This backend service is referenced when creating the forwarding rules to pass traffic for inspection to the FortiGate-VMs.

```
gcloud compute backend-services add-backend fgt-nsi-us-west1-lb \
    --region=us-west1 \
    --instance-group=fgt-nsi-us-west1-multi-ig \
    --instance-group-region=us-west1
```

To create forwarding rules:

There is a forwarding rule for each zone of the FortiGate inspection. They are referenced when creating the NSI deployment group.

```
gcloud compute forwarding-rules create fgt-us-west1a \
    --region=us-west1 \
    --load-balancing-scheme=INTERNAL \
    --ip-protocol=UDP \
    --ports=6081 \
    --network-tier=PREMIUM \
    --backend-service=fgt-nsi-us-west1-lb \
    --subnet=fgt-nsi-west

gcloud compute forwarding-rules create fgt-us-west1b \
    --region=us-west1 \
    --load-balancing-scheme=INTERNAL \
    --ip-protocol=UDP \
    --ports=6081 \
    --network-tier=PREMIUM \
    --backend-service=fgt-nsi-us-west1-lb \
    --subnet=fgt-nsi-west

gcloud compute forwarding-rules create fgt-us-west1c1 \
    --region=us-west1 \
    --load-balancing-scheme=INTERNAL \
    --ip-protocol=UDP \
    --ports=6081 \
    --network-tier=PREMIUM \
    --backend-service=fgt-nsi-us-west1-lb \
    --subnet=fgt-nsi-west
```

Configuring resources to enable NSI

To enable and use the Google Cloud NSI feature with the previously created FortiGate-VMs, instances groups, and networking infrastructure you need create the intercept deployment group with the following configurations.

To create the intercept deployment group:

```
gcloud beta network-security intercept-deployment-groups create newfgt-nsi-ftnt-dg \
--location=global \
--project=<your project id> \
--network=fgt-nsi-ib-new \
--no-async
```

To create the intercept deployments in each zone and add it to the deployment group:

```
gcloud beta network-security intercept-deployments create fgt-nsi-us-west1a \
--location=us-west1-a \
--project=<your project id> \
--forwarding-rule=fgt-us-west1a \
--intercept-deployment-group=projects/<your project
id>/locations/global/interceptDeploymentGroups/newfgt-nsi-ftnt-dg \
--forwarding-rule-location=us-west1 \
--no-async

gcloud beta network-security intercept-deployments create fgt-nsi-us-west1b \
--location=us-west1-b \
--project=<your project id> \
--forwarding-rule=fgt-us-west1b \
--intercept-deployment-group=projects/<your project
id>/locations/global/interceptDeploymentGroups/newfgt-nsi-ftnt-dg \
--forwarding-rule-location=us-west1 \
--no-async

gcloud beta network-security intercept-deployments create fgt-nsi-us-west1c1 \
--location=us-west1-c \
--project=<your project id> \
--forwarding-rule=fgt-us-west1c1 \
--intercept-deployment-group=projects/<your project
id>/locations/global/interceptDeploymentGroups/newfgt-nsi-ftnt-dg \
--forwarding-rule-location=us-west1 \
--no-async
```

To create the intercept endpoints group on the consumer side:

This step references the intercept deployment group that you created earlier.

```
gcloud beta network-security intercept-endpoint-groups create newfgt-nsi-ftnt-epg \
--intercept-deployment-group newfgt-nsi-ftnt-dg \
--project <your project id> \
--location global \
--no-async
```

To associate the intercept endpoint group with the web VPC:

```
gcloud beta network-security intercept-endpoint-group-associations create new-fgt-nsi-ftnt-
epg-assoc \
--intercept-endpoint-group newfgt-nsi-ftnt-epg \
--network fgt-nsi-ib-new-web \
--project <your project id> \
--location global \
--no-async
```

To create a security profile with custom intercept:

The security profile and security profile group are used to propagate policy contents into the first party Google firewall (Cloud NGFW Enterprise). This step references the intercept or NSI endpoint created earlier.

```
gcloud beta network-security security-profiles custom-intercept create newfgt-nsi-ftnt-sp1 \
--intercept-endpoint-group newfgt-nsi-ftnt-epg \
--billing-project <your project id> \
--organization <your organization id> \
--location global
```

To create a security profile group with the intercept profile:

```
gcloud beta network-security security-profile-groups create newfgt-nsi-ftnt-spg1 \
--custom-intercept-profile newfgt-nsi-ftnt-sp1 \
--billing-project <your project id> \
--organization <your organization id> \
--location global
```

To create a firewall policy:

```
gcloud compute network-firewall-policies create newfgt-nsi \
--project <your project id>\
--global
```

To create ingress and egress firewall policy rules:

After the security profile group has been created, a firewall policy is used to decide which traffic should be inspected from that group. This section references the security profiles and groups created earlier. In addition, there is a new GCP firewall policy rule action: `apply_security_profile_group`. This action specifies that the traffic should be sent to the firewall endpoint for inspection in any zone and VPC that has a firewall endpoint associated. This is a terminal rule, meaning that once the traffic is intercepted it is basically allowed unless the intercepting FortiGate-VM rejects it.

```
gcloud beta compute network-firewall-policies rules create 10 \
--action=APPLY_SECURITY_PROFILE_GROUP \
--firewall-policy newfgt-nsi \
--global-firewall-policy \
--security-profile-group
organizations/769491663990/locations/global/securityProfileGroups/newfgt-nsi-ftnt-spg1 \
--layer4-configs all \
--src-ip-ranges 0.0.0.0/0 \
--dest-ip-ranges 0.0.0.0/0 \
--direction INGRESS

gcloud beta compute network-firewall-policies rules create 11 \
--action=APPLY_SECURITY_PROFILE_GROUP \
--firewall-policy newfgt-nsi \
--global-firewall-policy \
--security-profile-group
organizations/769491663990/locations/global/securityProfileGroups/newfgt-nsi-ftnt-spg1 \
--layer4-configs all \
--src-ip-ranges 0.0.0.0/0 \
--dest-ip-ranges 0.0.0.0/0 \
--direction EGRESS
```

To create firewall policy association for the webserver VPC:

```
gcloud compute network-firewall-policies associations create \
--name newfgt-nsi-policy-assoc \
--global-firewall-policy \
--firewall-policy newfgt-nsi \
--network fgt-nsi-ib-new-web \
--project <your project id>
```

To create webserver VPC Windows Servers in three zones:

```
gcloud compute instances create fgt-nsi-web-us-west1a
--project=<your project id>
--zone=us-west1-a
--machine-type=e2-medium
--network-interface=network-tier=PREMIUM,stack-type=IPV4_ONLY,subnet=fgt-nsi-web1-west
--metadata=<Your windows key here>
--create-disk=auto-delete=yes,boot=yes,image=projects/windows-cloud/global/images/windows-
server-2025-dc-v20250312,mode=rw,size=50,type=pd-balanced
--no-shielded-secure-boot
--reservation-affinity=any

gcloud compute instances create fgt-nsi-web-us-west1b
--project=<your project id>
--zone=us-west1-b
--machine-type=e2-medium
--network-interface=network-tier=PREMIUM,stack-type=IPV4_ONLY,subnet=fgt-nsi-web1-west --
metadata= <Your windows key here>
--create-disk=auto-delete=yes,boot=yes,image=projects/windows-cloud/global/images/windows-
server-2025-dc-v20250312,mode=rw,size=50,type=pd-balanced
--no-shielded-secure-boot
--reservation-affinity=any
```

Configuring the FortiGate-VM

The following FortiGate configuration is used as a reference and works within the example document environment. Edit the values for IP addresses used in this example for interface configurations, static routing entry with the values used from your environment and deployment of this feature.

To configure FortiOS global settings:

```
config system global
    set admintimeout 480
    set alias "doc-nsi"
end

config system geneve
    edit "gcp"
        set interface "port1"
        set type ppp
        set remote-ip 10.50.160.1
```

```
    next
end
```

To configure network interfaces:

You must increase the MTU setting to handle the additional bytes of the Geneve header. In the example, MTU is set to 1600 to accommodate the Geneve header.

```
config system interface
    edit "port1"
        set vdom "root"
        set mode dhcp
        set allowaccess ping https ssh http probe-response
        set type physical
        set mtu-override enable
        set mtu 1600 <- Jumbo Frame settings
    next
    edit "port2"
        set vdom "root"
        set vrf 5
        set allowaccess ping https ssh snmp http telnet radius-acct probe-response fabric
ftm speed-test
        set type physical
        set mtu-override enable
        set mtu 1460
    next
    edit "gcp"
        set vdom "root"
        set type geneve
        set snmp-index 9
        set interface "port1"
        set mtu-override enable
        set mtu 1522
    next
end
```

To configure static routes:

```
config router static
    edit 1
        set gateway 10.50.180.1
        set device "port2"
    next
    edit 2
        set dst 130.211.0.0 255.255.252.0
        set gateway 10.50.160.1
        set device "port1"
    next
    edit 3
        set dst 35.191.0.0 255.255.0.0
        set gateway 10.50.160.1
        set device "port1"
    next
    edit 4
        set distance 5
        set device "gcp"
```

```
    next
    edit 5
        set gateway 10.50.160.1
        set device "port1"
    next
end
```

To configure router policy:

```
config router policy
    edit 1
        set input-device "port1"
        set srcaddr "all"
        set dstaddr "all"
        set gateway 10.50.160.1
        set output-device "port1"
    next
    edit 2
        set input-device "gcp"
        set srcaddr "all"
        set dstaddr "all"
        set output-device "gcp"
    next
end
```

To configure Web Filter profile:

```
config webfilter profile
    edit "plpr"
        set comment "Default web filtering."
        config ftgd-wf
            unset options
            config filters
                edit 1
                    set category 1
                    set action block
                next
                edit 2
                    set category 36
                    set action block
                next
                edit 3
                    set category 37
                    set action block
                next
                edit 4
                    set category 30
                    set action block
                next
                edit 5
                    set category 14
                    set action block
                next
            end
        end
        set log-all-url enable
```

```
    next
end
```

To configure network policies:

```
config firewall ssl-ssh-profile
    edit "custom-cert"
        set comment "Read-only SSL handshake inspection profile."
        config https
            set ports 443
            set status certificate-inspection
            set quic bypass
        end
        config ftps
            set status disable
        end
        config imaps
            set status disable
        end
        config pop3s
            set status disable
        end
        config smtps
            set status disable
        end
        config ssh
            set ports 22
            set status disable
        end
        config dot
            set status disable
            set quic inspect
        end
    next
end

config firewall policy
    edit 3
        set name "test"
        set srcintf "gcp"
        set dstintf "gcp"
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "PING"
        set logtraffic disable
    next

    edit 4
        set name "genevepolicy"
        set srcintf "gcp"
        set dstintf "gcp"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
```

```
        set service "ALL"
        set utm-status enable
        set ssl-ssh-profile "custom-cert"
        set webfilter-profile "plpr"
        set logtraffic all
    next
end
```

Validating the configuration

To validate that the GCP forwarding rule sends traffic to the FortiGate from the GCP network via the forwarding rule and back out to the GCP network:

```
FortiGate-VM # diagnose sniffer packet any 'icmp and host 8.8.8.8' 4
 Using Original Sniffing Mode
 interfaces=[any]
 filters=[icmp and host 8.8.8.8]
2.933258 gcp in 10.12.0.44 -> 8.8.8.8: icmp: echo request
2.933258 gcp in 10.12.0.44 -> 8.8.8.8: icmp: echo request
2.933405 gcp out 10.12.0.44 -> 8.8.8.8: icmp: echo request
2.933405 gcp out 10.12.0.44 -> 8.8.8.8: icmp: echo request
2.939409 gcp in 8.8.8.8 -> 10.12.0.44: icmp: echo reply
2.939409 gcp in 8.8.8.8 -> 10.12.0.44: icmp: echo reply
2.939422 gcp out 8.8.8.8 -> 10.12.0.44: icmp: echo reply
2.939422 gcp out 8.8.8.8 -> 10.12.0.44: icmp: echo reply
```

To validate that the FortiGate de-encapsulates the Geneve tunnel via a sniffer running on the FortiGate-VM:

```
FortiGate-VM # diagnose sniffer packet any 'port 6081' 4
 Using Original Sniffing Mode
 interfaces=[any]
 filters=[port 6081]0.048347 port1 in 10.50.160.1.49143 -> 10.50.160.201.6081: udp 100
0.048347 port1 in 10.50.160.1.49143 -> 10.50.160.201.6081: udp 100
0.048380 port1 out 10.50.160.199.49143 -> 10.50.160.1.6081: udp 100
0.048380 port1 out 10.50.160.199.49143 -> 10.50.160.1.6081: udp 100
0.310770 port1 in 10.50.160.1.31617 -> 10.50.160.201.6081: udp 100
0.310770 port1 in 10.50.160.1.31617 -> 10.50.160.201.6081: udp 100
```

To validate web filter logs via the FortiOS CLI for blocked traffic and allowed traffic:

```
FortiGate-VM # execute log filter device 0
FortiGate-VM # execute log filter category 3
FortiGate-VM # execute log display
1: date=2025-04-08 time=11:20:11 eventtime=1744136411182667111 tz="-0700" logid="0318012800"
type="utm" subtype="webfilter" eventtype="ftgd_err" level="error" vd="root" policyid=4
poluuid="d668a0fa-09bb-51f0-b476-0cb43fa7befc" policytype="policy" sessionid=11665608
srcip=10.50.160.5 srcport=39020 srccountry="Reserved" srcintf="gcp" srcintfrole="undefined"
srcuuid="7fe1f118-0ff2-51f0-9f24-98b5c709208a" dstip=151.101.194.133 dstport=443
dstcountry="Japan" dstintf="gcp" dstintfrole="undefined" dstuuid="7fe1f118-0ff2-51f0-9f24-
98b5c709208a" proto=6 httpmethod="GET" service="HTTPS" hostname="filestore.fortinet.com"
agent="Wget/1.21.3" profile="plpr" action="blocked" reqtype="direct"
```

```
url="https://filestore.fortinet.com/fortiguard/test-files/ml/ai_sample1" sentbyte=172
rcvdbyte=0 direction="outgoing" msg="A rating error occurs" error="all Fortiguard servers
failed to respond"
2: date=2025-04-08 time=11:10:20 eventtime=1744135820087372013 tz="-0700" logid="0318012800"
type="utm" subtype="webfilter" eventtype="ftgd_err" level="error" vd="root" policyid=4
poluuid="d668a0fa-09bb-51f0-b476-0cb43fa7befc" policytype="policy" sessionid=11659435
srcip=10.50.160.5 srcport=48496 srccountry="Reserved" srcintf="gcp" srcintfrole="undefined"
srcuuid="7fe1f118-0ff2-51f0-9f24-98b5c709208a" dstip=151.101.194.133 dstport=80
dstcountry="United States" dstintf="gcp" dstintfrole="undefined" dstuuid="7fe1f118-0ff2-
51f0-9f24-98b5c709208a" proto=6 httpmethod="GET" service="HTTP" hostname="asu.edu"
agent="Wget/1.21.3" profile="plpr" action="blocked" reqtype="direct" url="http://asu.edu/"
sentbyte=122 rcvdbyte=0 direction="outgoing" msg="A rating error occurs" error="all
Fortiguard servers failed to respond"
3: date=2025-04-08 time=11:09:08 eventtime=1744135748915697392 tz="-0700" logid="0318012800"
type="utm" subtype="webfilter" eventtype="ftgd_err" level="error" vd="root" policyid=4
poluuid="d668a0fa-09bb-51f0-b476-0cb43fa7befc" policytype="policy" sessionid=11658690
srcip=10.50.160.5 srcport=37112 srccountry="Reserved" srcintf="gcp" srcintfrole="undefined"
srcuuid="7fe1f118-0ff2-51f0-9f24-98b5c709208a" dstip=154.52.16.195 dstport=443
dstcountry="Singapore" dstintf="gcp" dstintfrole="undefined" dstuuid="7fe1f118-0ff2-51f0-
9f24-98b5c709208a" proto=6 httpmethod="GET" service="HTTPS"
hostname="filestore.fortinet.com" agent="Wget/1.21.3" profile="plpr" action="blocked"
reqtype="direct" url="https://filestore.fortinet.com/fortiguard/test-files/ml/ai_sample1"
sentbyte=172 rcvdbyte=0 direction="outgoing" msg="A rating error occurs" error="all
Fortiguard servers failed to respond"
4: date=2025-04-08 time=11:04:47 eventtime=1744135486992286367 tz="-0700" logid="0318012800"
type="utm" subtype="webfilter" eventtype="ftgd_err" level="error" vd="root" policyid=4
poluuid="d668a0fa-09bb-51f0-b476-0cb43fa7befc" policytype="policy" sessionid=11655901
srcip=10.50.160.5 srcport=55116 srccountry="Reserved" srcintf="gcp" srcintfrole="undefined"
srcuuid="7fe1f118-0ff2-51f0-9f24-98b5c709208a" dstip=154.52.16.195 dstport=443
dstcountry="Singapore" dstintf="gcp" dstintfrole="undefined" dstuuid="7fe1f118-0ff2-51f0-
9f24-98b5c709208a" proto=6 httpmethod="GET" service="HTTPS"
hostname="filestore.fortinet.com" agent="Wget/1.21.3" profile="plpr" action="blocked"
reqtype="direct" url="https://filestore.fortinet.com/fortiguard/test-files/ml/ai_sample1"
sentbyte=172 rcvdbyte=0 direction="outgoing" msg="A rating error occurs" error="all
Fortiguard servers failed to respond"
5: date=2025-04-07 time=22:00:38 eventtime=1744088439000382205 tz="-0700" logid="0317013312"
type="utm" subtype="webfilter" eventtype="ftgd_allow" level="notice" vd="root" policyid=4
poluuid="d668a0fa-09bb-51f0-b476-0cb43fa7befc" policytype="policy" sessionid=11191706
srcip=10.50.160.5 srcport=56366 srccountry="Reserved" srcintf="gcp" srcintfrole="undefined"
srcuuid="7fe1f118-0ff2-51f0-9f24-98b5c709208a" dstip=154.52.16.195 dstport=443
dstcountry="Japan" dstintf="gcp" dstintfrole="undefined" dstuuid="7fe1f118-0ff2-51f0-9f24-
98b5c709208a" proto=6 httpmethod="GET" service="HTTPS" hostname="filestore.fortinet.com"
agent="Wget/1.21.3" profile="plpr" action="passthrough" reqtype="direct"
url="https://filestore.fortinet.com/fortiguard/test-files/ml/ai_sample1" sentbyte=229
rcvdbyte=0 direction="outgoing" msg="URL belongs to an allowed category in policy"
ratemethod="domain" cat=52 catdesc="Information Technology"
```

Appendix

Configuring a multiregion network for data or traffic inspection

To create the GCP VPC network:

```
gcloud compute networks create fgt-nsi-ib-new \
  --subnet-mode custom \
  --description="NSI data or traffic VPC network with regional subnets"
```

To create a subnet in us-central1:

```
gcloud compute networks subnets create fgt-nsi-central \
  --network=fgt-nsi-ib-new \
  --region=us-central1 \
  --range=10.50.150.0/24 \
  --enable-private-ip-google-access \
  --description="Data or Traffic Subnet in us-central1"
```

To create a subnet in us-east1:

```
gcloud compute networks subnets create fgt-nsi-east \
  --network=fgt-nsi-ib-new \
  --region=us-east1 \
  --range=10.50.155.0/24 \
  --enable-private-ip-google-access \
  --description="Data or Traffic Subnet in us-east1"
```

To create a subnet in us-west1:

```
gcloud compute networks subnets create fgt-nsi-west \
  --network=fgt-nsi-ib-new \
  --region=us-west1 \
  --range=10.50.160.0/24 \
  --enable-private-ip-google-access \
  --description="Data or Traffic Subnet in us-west1"
```

Creating a multiregion management VPC with subnets

To create a multiregion management VPC with subnets:

```
gcloud compute networks create fgt-nsi-ib-new-mgmt \
  --subnet-mode=custom \
  --description="My FortiGate management VPC network with regional subnets"
```

To create a management subnet in us-central1:

```
gcloud compute networks subnets create fgt-nsi1-central-mgmt \
  --network=fgt-nsi-ib-new-mgmt \
  --region=us-central1 \
  --range=10.50.170.0/24 \
```

```
  --enable-private-ip-google-access \
  --description="My FortiGate management Subnet in us-central1"
```

To create a management subnet in us-east1:

```
gcloud compute networks subnets create fgt-nsi1-mgmt-east \
  --network=fgt-nsi-ib-new-mgmt \
  --region=us-east1 \
  --range=10.50.175.0/24 \
  --enable-private-ip-google-access \
  --description="My FortiGate management Subnet in us-east1"
```

To create a management subnet in us-west1:

```
gcloud compute networks subnets create fgt-nsi1-mgmt-west \
  --network=fgt-nsi-ib-new-mgmt \
  --region=us-west1 \
  --range=10.50.180.0/24 \
  --enable-private-ip-google-access
```

Creating a multiregion web VPC with subnets

To create a multiregion web VPC with subnets:

```
gcloud compute networks create fgt-nsi-ib-new-web \
  --subnet-mode custom \
  --description "My Public Web VPC network with regional subnets"
```

To create a subnet in us-central1:

```
gcloud compute networks subnets create fgt-nsi-web1-central1 \
  --network=fgt-nsi-ib-new-web \
  --region=us-central1 \
  --range=10.10.0.0/24 \
  --enable-private-ip-google-access \
  --description="Public Web Subnet in us-central1"
```

To create a subnet in us-east1:

```
gcloud compute networks subnets create fgt-nsi-web1-east1 \
  --network=fgt-nsi-ib-new-web \
  --region=us-east1 \
  --range=10.12.0.0/24 \
  --enable-private-ip-google-access \
  --description="Public Web Subnet in us-east1"
```

To create a subnet in us-west1:

```
gcloud compute networks subnets create fgt-nsi-web1-west \
  --network=fgt-nsi-ib-new-web \
  --region=us-west1 \
  --range=10.50.160.0/24 \
```

```
--enable-private-ip-google-access \
--description="Public Web Subnet in us-west1"
```